Machine Learning Fundamentals

Tags: ml, python, ai, optimization | Last Updated: Nov 05, 2026 22:33 UTC | DevNot

Gradient descent is an optimization algorithm used to minimize the loss function.

CODE (PYTHON)

```python
learning_rate = 0.01
gradient = 3.5
weight = 10.0
new_weight = weight - learning_rate * gradient
print(f'Updated Weight: {new_weight:.4f}')
```

OUTPUT (23.95ms)

```
Updated Weight: 9.9650
```

Notice how the parameter step reduces the loss proportionally to the gradient.

Python Decorators

A decorator is a callable that takes another function as an argument, extends its beh
avior without modifying it, and returns the modified function.

CODE (PYTHON)

```python
i = 10
while i < 1000:
    if i < 100:
        print(i)
    else:
        print("bye")
    i += 1  # Crucial: increment i to prevent an infinite loop
```

OUTPUT (44.94ms)

```
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
```

here we tried out the loop and the conditional code block

HANDWRITTEN NOTE: Handwritten Notes & Sketch



CODE (PYTHON)

```python
import functools
import time

def timer(func):
    @functools.wraps(func)
    def wrapper(*args, **kwargs):
        start = time.perf_counter()
        val = func(*args, **kwargs)
        print(f'{func.__name__} executed in {time.perf_counter() - start:.4f}s')
        return val
    return wrapper

@timer
def compute():
    return sum(i * i for i in range(100_000))

print('Result:', compute())
```

HANDWRITTEN NOTE: Loss Curve Graph

Continued editing after round-trip PDF import!